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WordPress Developer’s Intro To Docker, Part Three: Kubernetes

Yusuf Mudagal 28 Dec 2022 Updated 7 Aug 2026 3 min read

Welcome to the third and final part of our series on Docker for WordPress developers! In this installment, we’ll be introducing Kubernetes, the powerful container orchestration tool that allows you to manage and deploy your Docker containers at scale.

But before we dive into Kubernetes, let’s quickly review the basics of Docker and containerization.

What is Docker?

Docker is a tool designed to make it easier to create, deploy, and run applications by using containers. Containers allow a developer to package up an application with all of the parts it needs, such as libraries and other dependencies, and ship it all out as one package. By doing so, developers can rest assured that the application will run on any other Linux machine regardless of any customized settings that machine might have that could differ from the machine used for writing and testing the code.

What is Containerization?

Containerization is the process of packaging an application and its dependencies in a container to ensure that it can be easily moved from one environment to another. Containerization allows for consistent and reliable application deployment, regardless of the environment in which it is running.

Now that we’ve reviewed the basics of Docker and containerization, let’s move on to our main topic: Kubernetes.

What is Kubernetes?

Kubernetes, also known as K8s, is an open-source container orchestration system for automating the deployment, scaling, and management of containerized applications. It was originally developed by Google and is now maintained by the Cloud Native Computing Foundation (CNCF).

Kubernetes provides a platform-agnostic way to manage containerized applications, making it easier to deploy and manage applications in a variety of environments. It is designed to be highly scalable, allowing you to easily deploy and manage hundreds or even thousands of containers.

How Does Kubernetes Work?

Kubernetes works by managing a cluster of nodes, where each node is a worker machine in the cluster. These nodes run the applications and services that make up

your containerized application.

Kubernetes uses a system of pods to manage the containers on each node. A pod is a logical host for one or more containers, and it is the smallest deployable unit in Kubernetes. Pods are designed to be ephemeral, meaning that they can be created and destroyed as needed. This allows for flexible scaling of your application.

In addition to pods, Kubernetes also uses a system of controllers to manage the desired state of the cluster. These controllers, such as the Replication Controller and the Deployment Controller, ensure that the desired number of replicas of a particular pod are running at all times. If a pod goes down, the controller will automatically spin up a new one to replace it.

Benefits of Using Kubernetes

There are several benefits to using Kubernetes for managing your containerized applications:

- **Scalability:** Kubernetes makes it easy to scale your application up or down by adding or removing pods as needed.
- **Fault tolerance:** Kubernetes has built-in mechanisms for handling failures, such as automatically replacing failed pods to ensure that your application stays up and running.
- **Easy deployment:** Kubernetes provides a declarative approach to deployment, allowing you to specify the desired state of your application and letting Kubernetes handle the details of actually deploying and managing it.
- **Portability:** Kubernetes is platform-agnostic, meaning that it can be used to manage applications running.

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